Point-by-Point Response to Reviewer Comments (Point-wise)

Title	**Forecasting Indonesian Coal Sector Stock Prices Using a Hybrid ARIMA-LSTM Framework**
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Dear Editor and Reviewers, Thank you for the careful evaluation and constructive comments. We have revised the manuscript thoroughly and addressed each suggestion point by point. The location references below are based on the clean revised manuscript, which renders to 25 pages. Line numbers are counted within the cited paragraph or manuscript element, rather than as absolute line numbers for the whole page.

Reviewer 1

No	Reviewer's comments/feedback	Author's response/revision
1	**Does the title reflect the content, specific and effective?** *Yes* Further Suggestions: 1. Consider making the title more specific by highlighting the hybrid forecasting approach and selected Indonesian coal stocks. 2. Clarify whether the main contribution is the forecasting model or the dashboard implementation. 3. Avoid titles that imply general investment prediction capability.	**Response:** **1.** The title was revised to "Forecasting Indonesian Coal Sector Stock Prices Using a Hybrid ARIMA-LSTM Framework." The new title emphasizes the ARIMA-LSTM forecasting contribution and the specific Indonesian coal-sector stocks, while removing the dashboard/information-system wording from the title.
2	**Abstracts and keywords have described the overall content?** *Yes* Further Suggestions: 1. Clarify that long-term projections are exploratory scenarios rather than investment recommendations. 2. Mention the evaluation setting and comparison strategy more clearly. 3. Avoid claiming model superiority without baseline comparison. 4. Improve keywords by adding terms related to financial forecasting and decision-support systems.	**Response:** **1.** The abstract now states that the hybrid model is compared with standalone ARIMA and standalone LSTM using the January 2023–December 2024 out-of-sample period and MAE, RMSE, MAPE, and R^2. **2.** Stock-specific results are reported without claiming universal hybrid superiority, including positive and negative R^2 outcomes across stocks. **3.** The 2025–2045 forecasts are explicitly described as exploratory methodological scenarios rather than investment recommendations. **4.** The keywords were expanded to include "financial forecasting" and "decision-support systems."
3	**Background issues, problem formulation and objectives research, when viewed from the clarity of the definition** *Enough* Further Suggestions for Introduction: 1. Strengthen the research gap by comparing this study with previous forecasting approaches. 2. Clearly explain the novelty of the proposed framework. 3. Reduce unrelated background discussion and focus more on financial forecasting challenges. 4. Add clear research objectives or research questions.	**Response:** **1.** The research gap was strengthened by comparing previous single-model and hybrid forecasting approaches and highlighting the limited evidence on whether hybrid-model performance remains consistent across stocks with different characteristics. The revised manuscript also identifies the underexplored integration of stock-specific model evaluation, exploratory long-term projections, and an analytical dashboard for Indonesian coal-sector stocks. *Location: pp. 2–3, Introduction, para. 4, continuing across the page break.* **2.** The novelty and contribution of the proposed framework were clarified by emphasizing the stock-specific evaluation of hybrid ARIMA-LSTM performance for ADRO, PTBA, INDY, and ITMG, comparison with standalone ARIMA and LSTM baselines, and the integration of exploratory long-term projections with a web-based analytical interface. The manuscript further emphasizes that hybrid superiority should be evaluated rather than assumed. *Location: p. 3, Introduction, final paragraph, especially lines 6–9.* **3.** Unrelated background discussion was reduced and the Introduction was refocused on Indonesian coal-stock characteristics, financial time-series forecasting challenges, linear and nonlinear price behavior, and limitations of single-model forecasting approaches. *Location: p. 2, Introduction, paras. 1–3.* **4.** A clear research objective was explicitly added. The study now states that its objective is to develop and evaluate a hybrid ARIMA-LSTM framework for forecasting the stock prices of ADRO, PTBA, INDY, and ITMG, followed by baseline comparison, stock-specific evaluation, exploratory projections, and dashboard implementation. *Location: p. 3, Introduction, final paragraph, lines 1–9.*

4	**Research method used, when viewed from the formulation problem and purpose:** *Poor* Further Suggestions for Methods: 1. Provide more information about dataset preparation and model configuration. 2. Explain the ARIMA parameter-selection process more clearly. 3. Clarify how LSTM parameters were selected. 4. Report implementation details to improve reproducibility. 5. Explain why only historical closing prices were used without external market variables.	**Response:** **1.** Dataset preparation was expanded to report Yahoo Finance/yfinance, tickers, adjusted closing prices, company-specific observation periods, chronological train/test splitting, and the 15% LSTM validation split. *Location: pp. 4–5, Section 2.2, paras. 1–4 and Table 1; p. 4 para. 1 lines 1–13; p. 5 paras. 2–4.* **2.** Preprocessing was expanded to describe missing-value handling, chronological ordering, return calculation, ADF/KPSS testing, first-order differencing, Min-Max scaling, leakage prevention, and 30-observation sliding windows. *Location: pp. 5–6, Section 2.3, paras. 1–4; p. 6 lines 1–25.* **3.** ARIMA parameter selection was clarified: d = 1 is based on stationarity testing, while p and q are searched from 0 to 4 and selected using the minimum AIC. Company-specific orders are reported in Table 2. *Location: p. 7, Section 2.4, paras. 3–4, especially para. 4 lines 1–7; Table 2.* **4.** LSTM hyperparameters were explicitly described as fixed across all four stocks; the validation subset is used for early stopping and adaptive learning-rate reduction rather than hyperparameter search. The full architecture and training configuration are reported in Table 3. *Location: pp. 8–9, Section 2.5, para. 3, lines 1–10; Table 3.* **5.** Reproducibility details were added, including Python, yfinance, statistical/machine-learning libraries, TensorFlow/Keras, and a fixed random seed of 2024. *Location: p. 8, Section 2.5, para. 3, lines 7–10.* **6.** The univariate historical-price-only design is justified and its limitation is stated; coal prices, exchange rates, macroeconomic indicators, volume, and fundamentals are identified as excluded variables for future multivariate work. *Location: p. 5, Section 2.2, para. 3, lines 1–6.*
5	**Clarity and comprehensiveness of methods / analysis and synthesis with the results of the discussion, have you answered the problem thoroughly?** *No* Further Suggestions for Results: 1. Add comparison with individual ARIMA and LSTM models. 2. Provide additional validation to support the reliability of the results. 3. Include prediction results for all stocks, not only selected examples. 4. Explain the reasons behind different performances among companies. Further Suggestions for Discussion: 1. Discuss the impact of market volatility and structural changes. 2. Explain limitations of using only historical prices. 3. Interpret long-term forecasting results carefully because uncertainty increases over time. 4. Discuss future improvements such as external financial indicators and adaptive models.	**Response:** **1.** A direct out-of-sample comparison of ARIMA, standalone LSTM, and hybrid ARIMA-LSTM was added for all four stocks using MAE, RMSE, and MAPE. *Location: p. 14, Section 3.1.2(A), Table 6 and the following interpretation paragraph, lines 1–6.* **2.** R^2 was added as an additional evaluation metric for all three models, complementing the error-based metrics. *Location: pp. 14–15, Section 3.1.2(A), Table 7 and interpretation paragraph; p. 15 para. 1, lines 1–8.* **3.** Additional robustness testing was added through three-fold expanding-window validation using 2020, 2021, and 2022 as pre-test evaluation periods, while 2023–2024 remains the final holdout. *Location: pp. 10–11, Section 2.7, evaluation paragraph; p. 11 continuation lines 1–5; p. 15, Table 8 and validation paragraph, lines 1–6.* **4.** Prediction results for all four stocks were added as actual-versus-hybrid plots. *Location: p. 16, Section 3.1.2(B), para. 1, lines 1–5 and Figure 3.* **5.** The Discussion now explains stock-specific differences, including the weak ARIMA baseline for INDY, error propagation in the hybrid structure, and the regime-dependent results from expanding-window validation. *Location: p. 19, Section 3.2.1, paras. 1–3; para. 2 lines 1–7 and para. 3 lines 1–6.* **6.** Volatility, level shifts, structural breaks, commodity cycles, exchange rates, interest rates, regulation, geopolitical events, and firm-specific information are discussed as possible sources of forecast variation and error. *Location: pp. 19–20, Section 3.2.2, paras. 1–2; p. 20 para. 2, lines 1–5.* **7.** Long-term projections are presented with uncertainty bounds and interpreted as conditional/exploratory scenarios whose uncertainty increases with the horizon. *Location: pp. 17–18, Section 3.1.3(A), Table 9 and Figure 4; pp. 20–21, Section 3.2.4, paras. 1–2.* **8.** Future work now includes more rolling-origin windows, external financial indicators, multivariate forecasting, and more comprehensive uncertainty estimation. *Location: p. 22, Conclusion, final paragraph, lines 3–6.*
6	**The conclusions in this paper:** *Enough* Further Suggestions for Conclusion: 1. Clarify that the findings are limited to the evaluated stocks and period. 2. Avoid implying that the forecasting system can directly support investment decisions. 3. Add limitations and future research directions.	**Response:** **1.** The conclusion now emphasizes stock-specific performance and states that ARIMA-LSTM hybridization does not guarantee uniform improvement. *Location: p. 21, Section 4, para. 2, lines 1–6.* **2.** The long-term projections are explicitly described as exploratory scenarios, and the dashboard is positioned as an analytical interface rather than an investment-recommendation system. *Location: p. 21, Section 4, para. 3, lines 1–7.* **3.** The limitations are explicitly restricted to four stocks, historical closing-price inputs, the 2023–2024 final test period, and the selected three-fold expanding-window design; future work is also stated. *Location: pp. 21–22, Section 4, final paragraph; p. 22 lines 1–6.*

7	**Have references been written with citation tools?** *No* Further Suggestions for References: 1. Add recent journal references on hybrid forecasting and financial time-series prediction. 2. Include original references for ARIMA and LSTM. 3. Improve consistency in reference formatting.	**Response:** **1.** Recent international references on hybrid forecasting, financial forecasting, leakage-controlled evaluation, structural breaks, and model comparison were added and integrated into the Methods and Discussion. *Location: pp. 23–25, References [18], [27]–[34], with corresponding in-text citations in Sections 2.3–3.2.* **2.** Foundational references were added and positioned next to the relevant methodological descriptions: Box and Jenkins for ARIMA and Hochreiter and Schmidhuber for LSTM. *Location: p. 3, Section 2.1, para. 1, lines 5–6 for [12]; p. 8, Section 2.5, para. 1, lines 1–3 for [23]; reference entries [12] on p. 23 and [23] on p. 24.* **3.** Reference formatting and DOI presentation were standardized across the reference list. *Location: pp. 22–25, References [1]–[34].*

Reviewer 2

No	Reviewer's comments/feedback	Author's response/revision
1	**Does the title reflect the content, specific and effective?** *Yes* Further Suggestions: 1. Highlight the ARIMA-LSTM forecasting contribution. 2. Specify that the study focuses on Indonesian coal-sector stocks.	**Response:** **1.** The title was revised to emphasize both the hybrid ARIMA-LSTM forecasting framework and its application to Indonesian coal-sector stocks. *Location: p. 1, title block, lines 1–2.*
2	**Abstracts and keywords have described the overall content?** *Yes* Further Suggestions: 1. Explain the evaluation procedure more clearly. 2. Clarify the meaning of long-term projections. 3. Avoid general claims about forecasting performance without sufficient comparison.	**Response:** **1.** The abstract now states the January 2023–December 2024 out-of-sample evaluation period, the standalone ARIMA/LSTM baselines, and the four evaluation metrics. *Location: p. 1, Abstract, para. 1, lines 1–15.* **2.** The abstract reports stock-specific results and explicitly notes that hybridization did not consistently improve performance. *Location: p. 1, Abstract, para. 1, lines 13–18.* **3.** The 2025–2045 projections are explicitly described as exploratory methodological scenarios rather than investment recommendations. *Location: p. 1, Abstract, para. 1, lines 18–20.*
3	**Background issues, problem formulation and objectives research, when viewed from the clarity of the definition** *Good* Further Suggestions for Introduction: 1. Provide stronger justification for selecting coal stocks. 2. Discuss previous forecasting limitations more clearly. 3. Explain the expected contribution of the proposed hybrid model.	**Response:** **1.** The rationale for selecting Indonesian coal-sector stocks was strengthened by linking their prices to commodity cycles, coal demand, regulation, economic uncertainty, and high volatility. *Location: p. 2, Introduction, para. 1, lines 1–8.* **2.** Previous forecasting limitations and the motivation for hybrid modelling are discussed through comparison with single-model approaches and prior hybrid studies. *Location: pp. 2–3, Introduction, paras. 3–4.* **3.** The expected contribution is explicitly stated as stock-specific model evaluation, baseline comparison, exploratory long-term projection, and dashboard presentation. *Location: p. 3, Introduction, para. 5, lines 1–9.*
4	**Research method used, when viewed from the formulation problem and purpose:** *Poor* Further Suggestions for Methods: 1. Explain data preprocessing and model training procedures more clearly. 2. Add baseline models for comparison. 3. Report important implementation details and parameter settings. 4. Include validation strategies beyond a single testing period.	**Response:** **1.** Data preprocessing and model training procedures were expanded, covering missing values, stationarity testing, differencing, scaling, sliding windows, ARIMA selection, LSTM configuration, and hybrid residual modelling. *Location: pp. 5–9, Sections 2.3–2.6.* **2.** Standalone ARIMA and standalone LSTM are explicitly included as baseline models under the same data split and evaluation procedure. *Location: p. 9, Section 2.6, final paragraph, lines 1–3.* **3.** Important LSTM settings and implementation details are now reported, including 30-day windows, two 64-unit LSTM layers, dropout 0.20, Dense 32/ReLU, Adam 0.001, MSE, batch size 32, 100 epochs, 15% validation, patience 15, TensorFlow/Keras, and random seed 2024. *Location: pp. 8–9, Section 2.5, para. 3 and Table 3.* **4.** Validation beyond a single holdout was added using three expanding-window pre-test folds for 2020, 2021, and 2022, while preserving 2023–2024 as the final holdout. *Location: pp. 10–11, Section 2.7, evaluation paragraph; p. 11 continuation lines 1–5; p. 15, Table 8.*
5	**Clarity and comprehensiveness of methods / analysis and synthesis with the results of the discussion, have you answered the problem thoroughly?** *No* Further Suggestions for Results: 1. Add more detailed performance comparison.	**Response:** **1.** The Results now provide a detailed three-model comparison using MAE, RMSE, and MAPE for all stocks. *Location: p. 14, Section 3.1.2(A), Table 6 and interpretation paragraph, lines 1–6.* **2.** R^2 was added for ARIMA, standalone LSTM, and hybrid ARIMA-LSTM as an additional evaluation metric. *Location: pp. 14–15, Table 7 and its*

	2. Include additional forecasting metrics. 3. Provide uncertainty analysis. 4. Report computational complexity if efficiency is discussed. Further Suggestions for Discussion: 1. Explain differences in stock behavior and volatility. 2. Discuss possible causes of forecasting errors. 3. Compare findings with recent forecasting studies. 4. Clearly state study limitations.	*interpretation paragraph; p. 15 lines 1–8.* **3.** Uncertainty analysis was added for the long-term projections. The Methods specify ARIMA/SARIMAX 95% confidence intervals, while the Results show lower/upper bounds and widening uncertainty over time. *Location: p. 11, Section 2.7, final paragraph, lines 1–7; pp. 17–18, Table 9 and Figure 4.* **4.** Computational complexity was not added because the manuscript does not make a computational-efficiency claim; this reviewer request was conditional ("if efficiency is discussed"). *Location: Not applicable; no efficiency claim is made in the revised manuscript.* **5.** The Discussion explains stock behavior, volatility, INDY's downward level shift, weak ARIMA baseline, residual error propagation, and regime-dependent performance. *Location: pp. 19–20, Sections 3.2.1–3.2.2.* **6.** Possible causes of forecasting errors were explicitly discussed. The INDY case demonstrates error propagation from a weak ARIMA baseline, while structural breaks, commodity-price cycles, exchange-rate movements, interest rates, regulation, geopolitical events, and firm-specific information are identified as possible sources of persistent forecasting error. *Location: p. 19, Section 3.2.1, para. 2; p. 20, Section 3.2.2, para. 2.* **7.** The findings are compared with recent hybrid forecasting studies and the manuscript explains why its stock-specific results qualify general claims of hybrid superiority. *Location: p. 20, Section 3.2.3, para. 1, lines 1–8.* **8.** Study limitations are stated in the conclusion, including the four-stock scope, historical-price-only input, the final testing period, and the selected validation design. *Location: pp. 21–22, Section 4, final paragraph; p. 22 lines 1–6.*
6	**The conclusions in this paper:** *Enough* Further Suggestions for Conclusion: 1. Emphasize that results depend on stock characteristics. 2. Mention limitations of the current forecasting setup. 3. Add future work related to multivariate forecasting and uncertainty estimation.	**Response:** **1.** The conclusion explicitly states that forecasting performance depends on stock characteristics and that hybridization does not consistently outperform standalone LSTM. *Location: p. 21, Section 4, para. 2, lines 1–6.* **2.** Current limitations are explicitly stated for the four stocks, historical-price-only input, 2023–2024 final testing period, and three-fold expanding-window validation design. *Location: pp. 21–22, Section 4, final paragraph; p. 22 lines 1–3.* **3.** Future work now includes more rolling-origin windows, multivariate forecasting with external financial variables, and more comprehensive uncertainty-estimation methods. *Location: p. 22, Section 4, final paragraph, lines 3–6.*
7	**Have references been written with citation tools?** *No* Further Suggestions for References: 1. Add more recent international journal references. 2. Include foundational forecasting references. 3. Check formatting and DOI consistency.	**Response:** **1.** Recent international journal references were added in the areas of financial forecasting, hybrid models, evaluation, structural breaks, and forecasting combinations. *Location: pp. 23–25, References [18], [27]–[34].* **2.** Foundational forecasting references were included: Box and Jenkins for ARIMA and Hochreiter and Schmidhuber for LSTM. *Location: Reference [12], p. 23; Reference [23], p. 24; corresponding in-text citations at p. 3, Section 2.1, and p. 8, Section 2.5.* **3.** Reference formatting and DOI presentation were checked and standardized throughout the reference list. *Location: pp. 22–25, References [1]–[34].*

We sincerely appreciate the Editor and Reviewers for their constructive feedback. The revisions have improved the manuscript's clarity, methodological transparency, validation, reproducibility, and interpretation.